

Introducción a la Ciencia de Datos

Maestría en Ciencias
de la Computación

Dr. Irvin Hussein López Nava

The slide features a white background with several small, semi-transparent blue squares scattered around the edges. There are dark blue squares at the top center, bottom left, and middle right. Lighter blue squares are located at the top right, middle left, bottom center, and bottom right.

Presentación del curso

Objetivo del curso

El **curso** busca proporcionar al estudiante **herramientas** que le permitan iniciar en el área de **ciencias de datos**, en particular, generar **modelos de aprendizaje** basados en grandes volúmenes de datos.

Pretende también que el estudiante entienda y maneje los métodos de **estadística** e **inteligencia artificial** más empleados.



Temario

Unidad 1

2

3

4

5

6

7

Introducción

Procesamiento
de datos

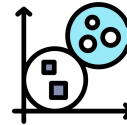
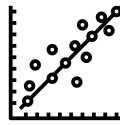
Regresión

Aprendizaje
supervisado

Aprendizaje no
supervisado

Meta
aprendizaje

Aprendizaje
profundo



Evaluación

20%

Lecturas y tareas



50%

Prácticas



30%

Proyecto final

- Las lecturas, tareas y prácticas son obligatorias y deben ser completadas en tiempo y forma.
- Por cada lectura deberá crearse un reporte de máximo una cuartilla.
- Para cada práctica debe crearse un Notebook y deberá agregarse al repositorio personal.
- El reporte del proyecto deberá describir el trabajo de investigación en formato de artículo.

Roadmap

13 semanas · 5 lecturas · 5 prácticas · 1 proyecto



Uso de la IA generativa

No está prohibida. Está delimitada y se declara.

Sí, adelante

- Depurar y explicar errores de tu código
- Explorar alternativas de implementación
- Aclarar conceptos vistos en clase
- Pulir la redacción de un texto que **tú** escribiste

No, aquí no

- Generar el reporte de lectura - el punto es tu lectura.
- Escribir el reporte o el análisis de las prácticas o proyecto
- Entregar código que no puedas explicar

La regla: declara en cada entrega qué herramienta fue usada y para qué.

Timeline del curso



Fingerspelling Recognition in Mexican Sign Language (LSM) Using Machine Learning

Ricardo Fernando Medina-Chavez¹, Jesús Javier González-Pérez², and Irvin Hussain López-Navu³

¹ Computer Science Department, Centro de Investigación Científica y de Educación Superior de Ensenada, 22060 Ensenada, BC, Mexico

² Facultad de Ciencias, Universidad Autónoma de Baja California,

22060 Ensenada, BC, Mexico

jesus.gonzalez@uabc.mx

Abstract. Sign languages allow deaf people to express their thoughts, emotions, and opinions in a complete and complete way, just like oral languages. Each sign language is unique and has its own grammar, syntax, and vocabulary. Mexican Sign Language (LSM) is characterized by rich gestural and facial expressions that give it a great communicative and linguistic capacity. In the study of LSM, two main components have been identified: (1) fingerspelling, and (2) ideograms. The first is similar to spelling in oral languages, and it is used to communicate proper names, technical terms or words for which there are no specific signs or which are little known to the deaf community. In this paper, we propose a method for recognizing the LSM alphabet by using machine learning-based techniques capable of classifying the signs made by 10 test subjects. 21 keypoints of the hands were extracted from the MediaPipe library in order to have a better representation to feed the classification models. The results when classifying the 21 letters resulted an F1 score of 0.98 with 3 of the 4 trained classifiers, and scoring values below 0.95 for less than 3 letters. Tools such as those proposed in this work can facilitate machine communication by translating Spanish into LSM and vice versa, allowing both communities to engage effectively in various settings.

Enhancing Keypoint Selection for Hand Model in Recognition of Mexican Sign Language Alphabet

Jesús Javier González-Pérez, Jesús Antonio Navarrete-López, Irvin Hussain López-Navu

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE),

Ensenada 22060, Baja California, México

{gonzarte, jnavarrete, hussain}@cicese.edu.mx

Abstract. The foundation of learning any language lies in its alphabet, which serves as the basis for word formation and expression. This principle also applies to sign languages, where the manual alphabet is primarily used to spell proper nouns, technical terms, and words without established signs. Technological advancements have enabled the tracking and interpretation of hand movements—particularly hand gestures—enabling the computational modeling of these alphabets. However, achieving efficient machine performance requires a challenge: refining the selection of keypoints for the Mexican Sign Language (LSM) alphabet by removing the most critical keypoints, simplifying the model, and while maintaining high performance (F1 score > 0.94). This bridge contributes to the development of lightweight and efficient systems for learning and displaying the LSM alphabet in real-time applications.

Keywords. Sign language recognition, Mexican sign language, LSM alphabet, hand landmarks, hand keypoints.

1 Introduction

Sign language recognition has emerged as a key area in computer vision, machine learning, and human-computer interaction, aiming to reduce communication barriers for Deaf communities. Mexican Sign Language (LSM), officially recognized in Mexico since 2005, is a full-fledged natural language with its own grammar, lexicon, and syntax [1]. One of its key components is the LSM manual alphabet, a set of 27 hand configurations representing the letters of the Spanish alphabet.

The LSM manual alphabet—commonly referred to as "fingerspelling"—is unique and has its own grammar, syntax, and vocabulary. Mexican Sign Language (LSM) is characterized by rich gestural and facial expressions that give it a great communicative and linguistic capacity. In the study of LSM, two main components have been identified: (1) fingerspelling, and (2) ideograms. The first is similar to spelling in oral languages, and it is used to communicate proper names, technical terms or words for which there are no specific signs or which are little known to the deaf community. In this paper, we propose a method for recognizing the LSM alphabet by using machine learning-based techniques capable of classifying the signs made by 10 test subjects. 21 keypoints of the hands were extracted from the MediaPipe library in order to have a better representation to feed the classification models. The results when classifying the 21 letters resulted an F1 score of 0.98 with 3 of the 4 trained classifiers, and scoring values below 0.95 for less than 3 letters. Tools such as those proposed in this work can facilitate machine communication by translating Spanish into LSM and vice versa, allowing both communities to engage effectively in various settings.

Linguistic studies show that fingerspelling in LSM is structured around questionnaires (hand shapes), grammars (inflections, location), and discourse (movement components) [2]. Each shape is the key parameter for distinguishing letters [3]. These signs are typically articulated near the chest and/or shoulder, providing clarity and visual accessibility. Most LSM alphabet signs are static with fixed handshapes, while a subset—including J, K, L, O, X, and Z—requires dynamic movements [3, 4]. Some signs are especially relevant for computational recognition systems due to their standardized form and consistent spatial occupation.

Recognition of LSM manual alphabet signs has been approached through various sensing modalities, each offering advantages for capturing handwriting intricacies. Early efforts primarily used sensor-based systems, such as non-invasive sensors [5] or high-resolution cameras [6]. More recent work has leveraged deep learning for high-level feature extraction and classification, showing promising results in controlled environments.



Satellite Imagery Classification Using Shallow and Deep Learning Approaches

Michelle Salas-Vinuet¹ and Irvin Hussain López-Navu²

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE),

22060 Ensenada, Baja California, Mexico

salas@cicese.edu.mx, hussain@cicese.mx

https://www.cicese.edu.mx/

Abstract. Recent advances in remote sensing technology and high-resolution satellite imagery offer great possibilities for understanding the earth's surface. However, satellite image classification is a challenging problem due to its high variability inherent in satellite data. For this purpose, two learning approaches are proposed and compared for classifying a large-scale dataset including different types of land-use and land-cover surfaces (Urban, Traditional (shallow) machine learning models and deep learning models are built by using a set of features extracted from the satellite images for both approaches and using the RGB images for deep models. The best F1 score obtained by the shallow approach was 0.87, while for the deep approach it was 0.91. No significant difference was found in these results; however, significant improvement can be made by exploring the deep approach in greater depth.

Keywords. Satellite imagery · Satellite imagery classification · Image classification · Land use classification · Land cover classification · Remote sensing · Deep learning

CHAPTER 2

Towards detection of depression and anxiety in adolescents in social media

Scarlett Magdaleno-García and Irvin Hussain López-Navu

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE),

22060 Ensenada, Baja California, Mexico

scarlett.mg@uabc.mx, hussain@cicese.mx

2.1. Introduction

Anxiety and depression are among the most prevalent mental health disorders globally, contributing significantly to the burden of disease and disability. According to recent data from the World Health Organization (WHO), anxiety disorders affect over 300 million people, while depression impacts more than 280 million individuals worldwide¹. Anxiety is characterized by an excessive or persistent sense of worry, fear, or unease. It often involves heightened physical arousal, such as rapid heart rate or sweating, and is associated with concerns about future events or outcomes. The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) categorizes anxiety disorders into various forms, such as Generalized Anxiety Disorder, Panic Disorder, and Social Anxiety Disorder. These disorders are distinguished by their ability to significantly interfere with daily functioning and cause marked distress.

Depression is a mood disorder characterized by persistent feelings of sadness, hopelessness, and a loss of interest or pleasure in previously enjoyed activities. It can also lead to physical symptoms, as well as cognitive impairments such as difficulty concentrating or making decisions. Major Depressive Disorder, as defined by the DSM-5, requires the presence of at least five symptoms (including either depressed mood or loss of interest) for a minimum of two weeks, which must represent a change from previous functioning and cause significant distress or impairment in social, occupational, or other important areas of life².

These conditions often co-occur [3], intensify each other's effects, and are associated with a wide range of negative outcomes, including increased risk of self-harm, impaired daily functioning, and increased risk of suicide. Despite their prevalence, underdiagnosis and

2021
60% inscritos

73% inscritos
2023

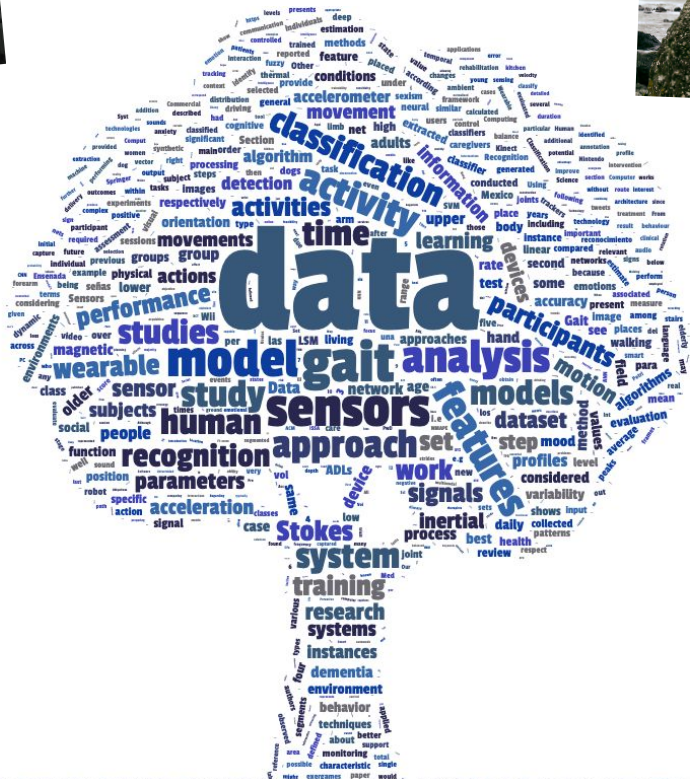
2024
66% inscritos

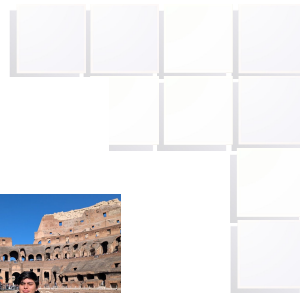
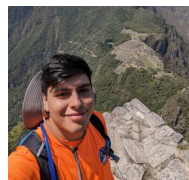
84% inscritos
2026

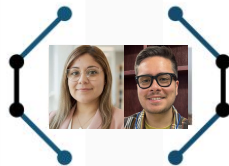
66% inscritos
2020

Secuencia de cursos









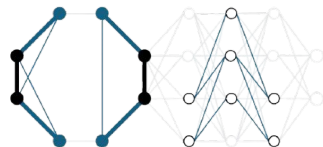
Classificación de actividad y comportamiento humano

Yulith V. Altamirano Flores
Sebastián Toro Guajardo



Detección temprana de somnolencia al conducir

Sarid García Pérez

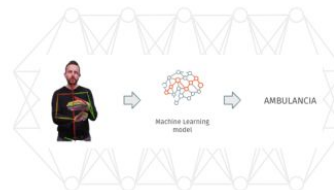


Laboratorio de Ciencia de Datos y Aprendizaje Automático



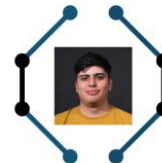
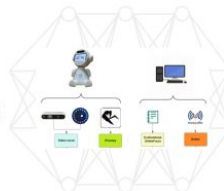
Traducción automática de la Lengua de Señas Mexicana

Michelle Sainos Vizuet
Ricardo F. Morfin Chávez
Jesús Antonio Navarrete López
Jorge Braulio Morales Martínez
Jesús Javier Gortarez Pelayo



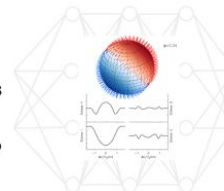
Comportamiento no verbal en interacción humano-robot

Ernesto A. Lozano De la Parra



Análisis de datos astronómicos

Joan M. Raygoza Romero

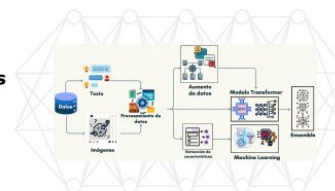


© 2025 Oleg Kochukhov
Original template design by Andreas Viklund



Análisis de publicaciones de redes sociales

Esteban Ponce León
Martha Paola Jiménez Martínez
Cielo Aholiva Higuera Gutiérrez



Monitoreo automático de especies marinas

Paola J. Delgado García
M. Camila López Posada



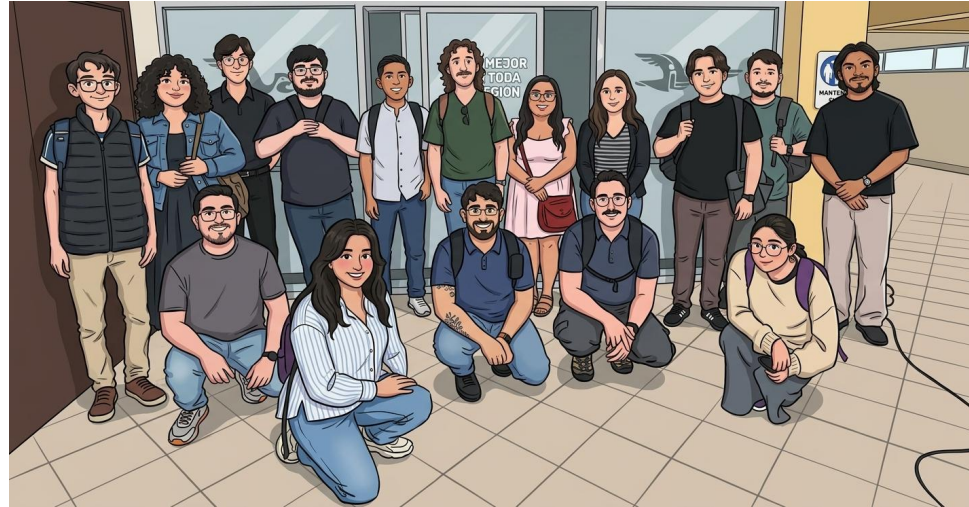
Medición y análisis de comportamiento canino

Eliaf Yahir García Loya



Acerca de ustedes

- Formación académica y procedencia
- Intereses de investigación
- Interés en el curso
- Algo extra-académico





01

Introducción



The slide features a white background with several dark blue squares of varying sizes scattered around the central text. There is one square at the top center, one at the top right, one at the bottom left, one at the bottom center, one at the bottom right, and one on the left side.

1.1 La Ciencia de Datos

[illegible]

¿Qué es la Ciencia de Datos?



Definiciones

ChatGPT

Campo interdisciplinario que involucra métodos, procesos, algoritmos y sistemas para obtener conocimiento y perspicacia a partir de datos en diversas formas

<https://chat.openai.com/>

Wikipedia

Campo académico interdisciplinario que utiliza estadística, computación científica, métodos, procesos, algoritmos y sistemas científicos para obtener, tratar, analizar y presentar informes a partir de datos ruidosos, estructurados y no estructurados.

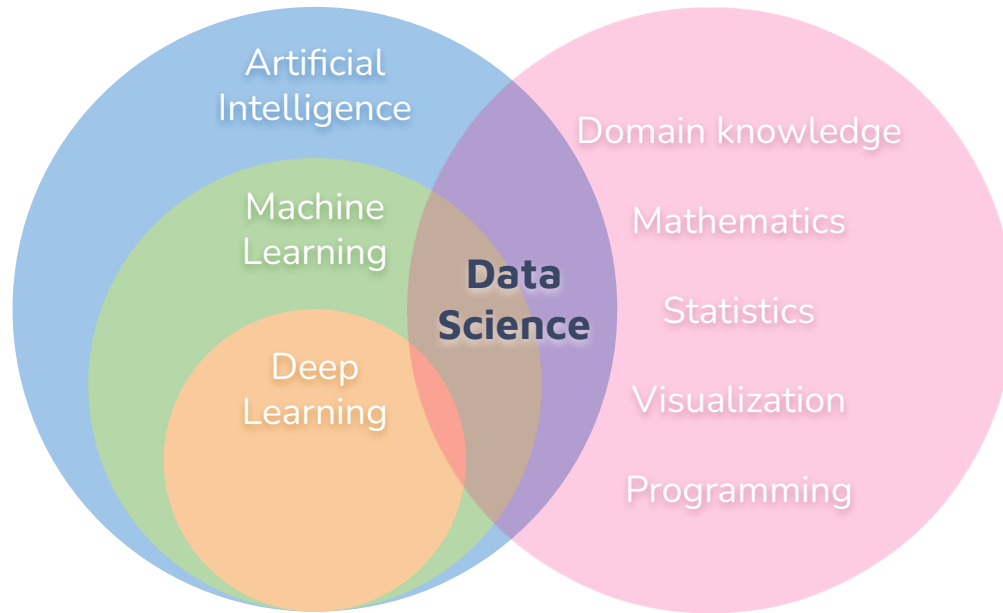
https://es.wikipedia.org/wiki/Ciencia_de_datos

Science Direct

La Ciencia de Datos proporciona procesos y sistemas para extraer y gestionar conocimientos a partir de fuentes de datos estructuradas y no estructuradas.

<https://www.sciencedirect.com/topics/social-sciences/data-science>

Campos relacionados



El término “Ciencia de Datos”

- El término “**ciencia**” implica **conocimiento** adquirido a través del **estudio sistemático**.
- En una definición: trabajo **sistemático** que construye y organiza el **conocimiento** en forma de explicaciones y predicciones.

Por lo tanto, la **ciencia de datos** podría implicar:

un enfoque que involucre **datos** y, por extensión, **estadística**;

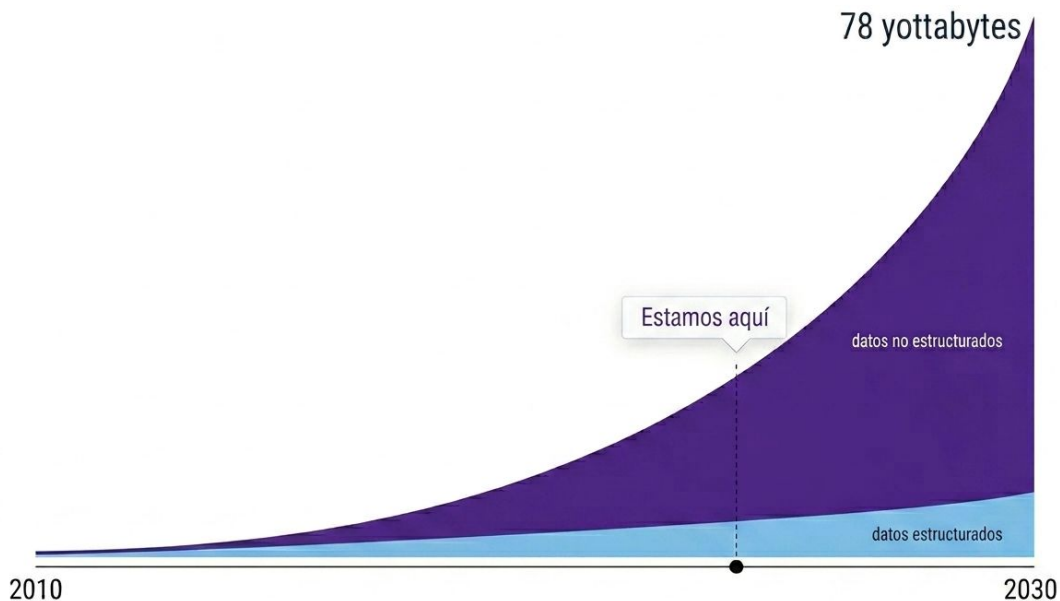
o el **estudio sistemático** de la organización, propiedades, y análisis de datos y su papel en la **inferencia**, incluida la confianza misma en la inferencia

¿Por qué el término “**ciencia de datos**”
cuando se ha tenido la “**estadística**”
durante siglos?



Necesidad de este término

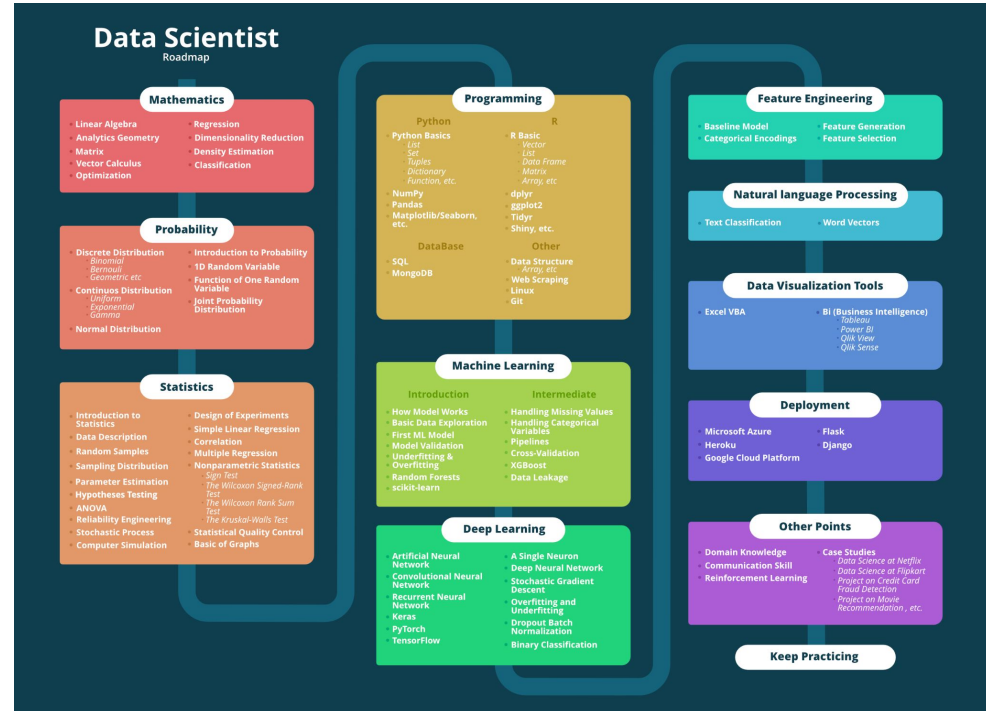
Primero: la materia prima, la parte de "**datos**", es cada vez más heterogénea y desestructurada: texto, imágenes, video...



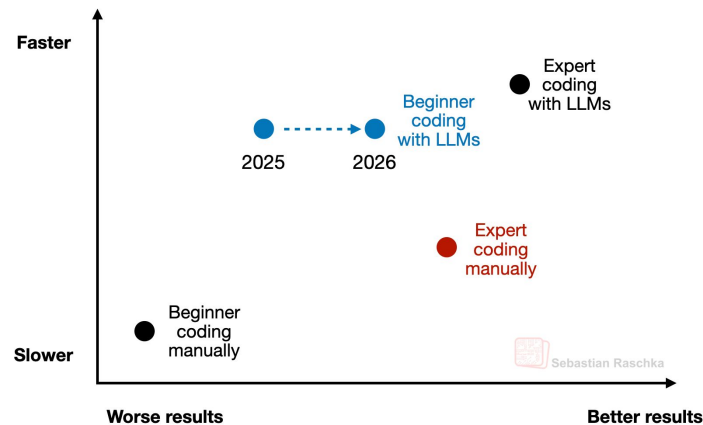
Necesidad de este término

Segundo: la proliferación de lenguajes diseñados para que las máquinas interpreten los **datos** automáticamente.

En otras palabras, las computadoras están haciendo cada vez más el trabajo de fondo.



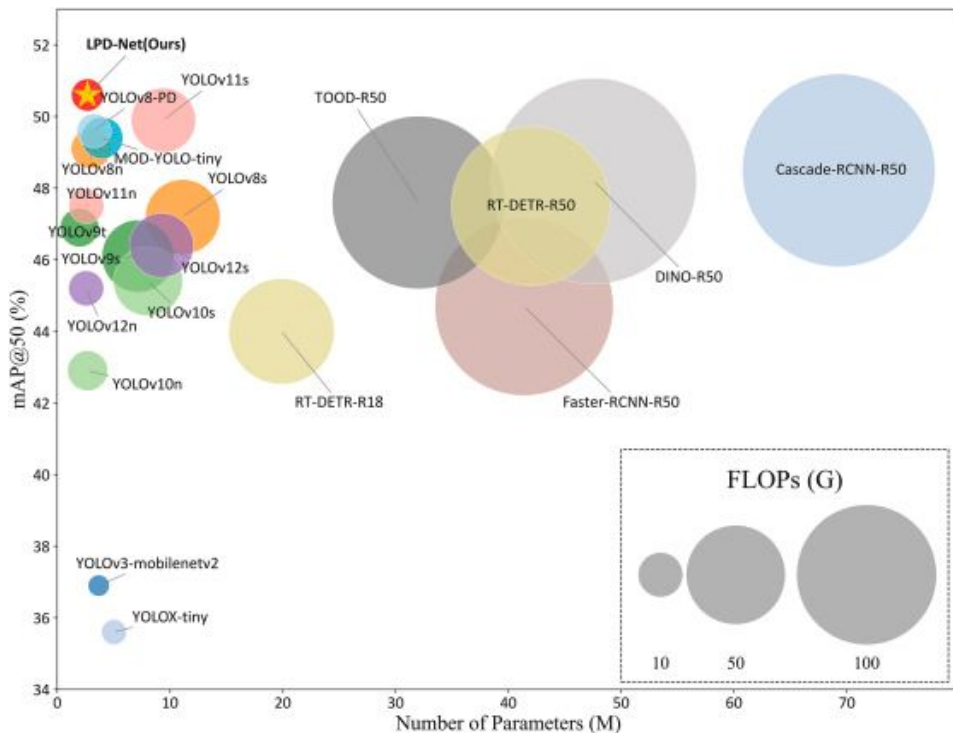
Necesidad de este término



Necesidad de este término

Tercero: el hardware de última generación que permite procesar grandes cantidades de información.

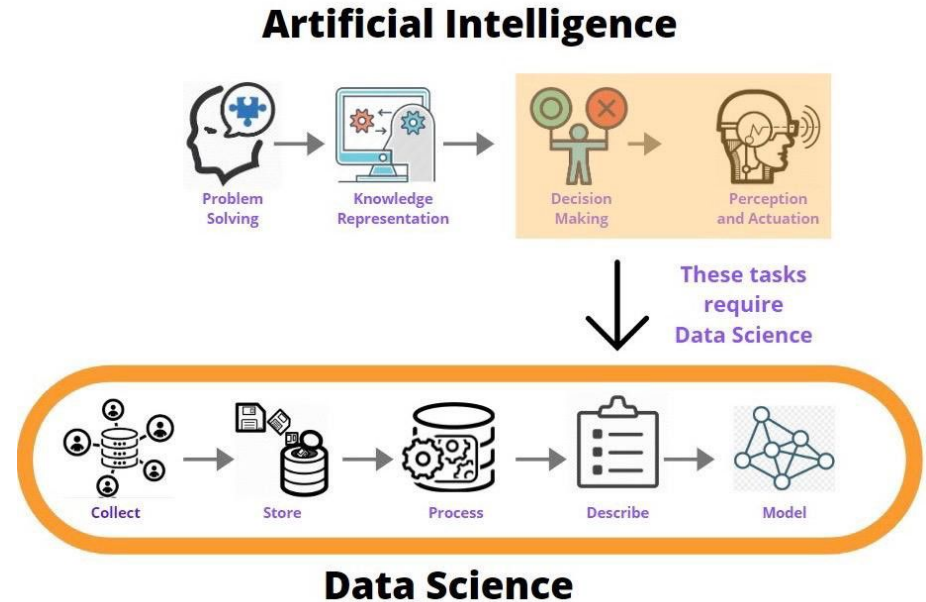
Algoritmos computacionales que fueron propuestos décadas atrás y que hoy en día pueden ser utilizados.



Necesidad de este término

Por último, la **toma de decisiones**:
Cada vez es más común que la computadora tome decisiones, sin la ayuda de humanos.

Sin embargo, hay una multitud de **problemas** que van desde los costos de decisiones incorrectas hasta problemas éticos y de privacidad.



El proceso de la Ciencia de Datos



Data Engineers

Data Analysts

Machine Learning Engineers

Data Scientists

Secuencia de cursos

1

Introducción a la
Ciencia de Datos

2

Ciencia de Datos para
Sensado Inteligente

3

Reconocimiento de
Patrones



¿Para qué se utiliza?

Análisis **descriptivo**

Examina los datos para obtener información sobre **lo que ha** ocurrido u ocurre en el entorno de los datos.

Análisis de **diagnóstico**

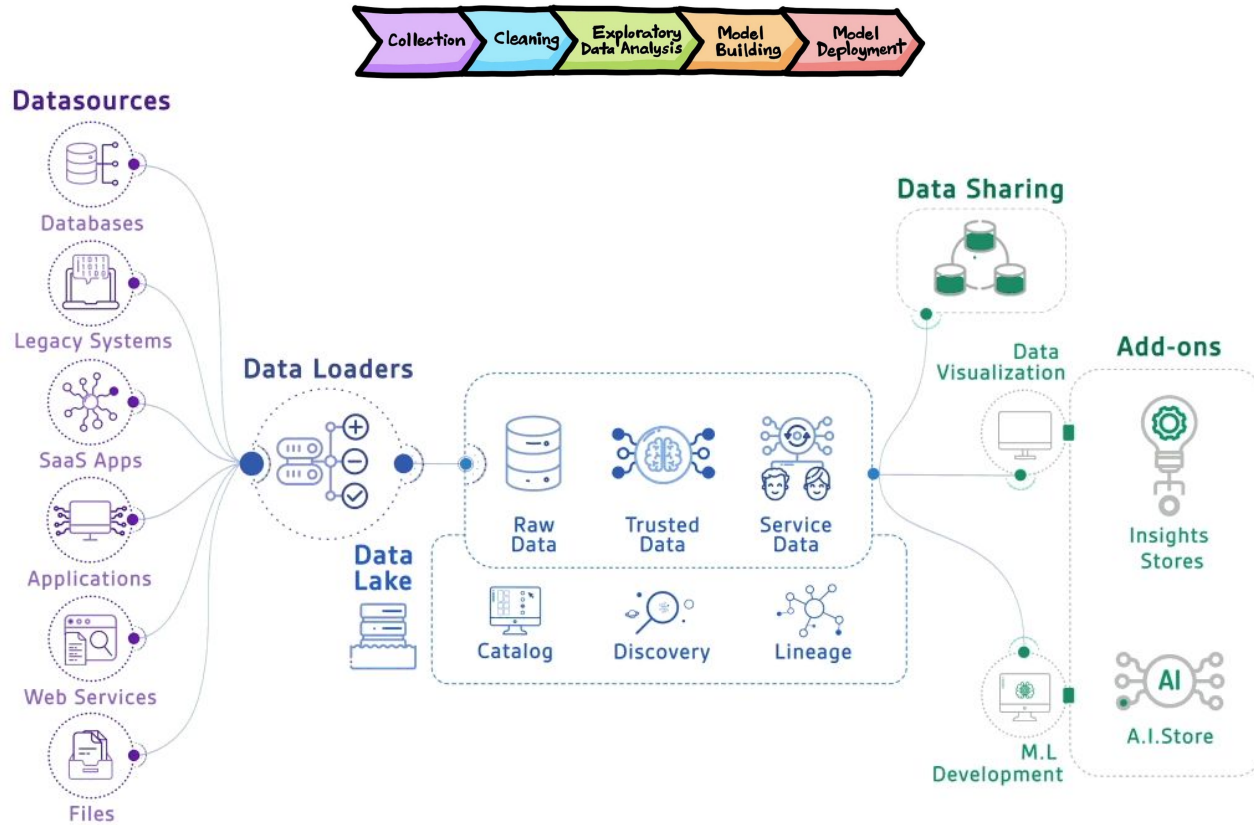
Es un análisis detallado de los datos para entender **por qué ha** ocurrido algo.

Análisis **predictivo**

Utiliza datos históricos para hacer **previsiones** sobre los patrones de datos que pueden producirse en el futuro.

Análisis **prescriptivo**

No solo predice lo que es probable que ocurra, sino que sugiere una respuesta **óptima** para ese resultado.

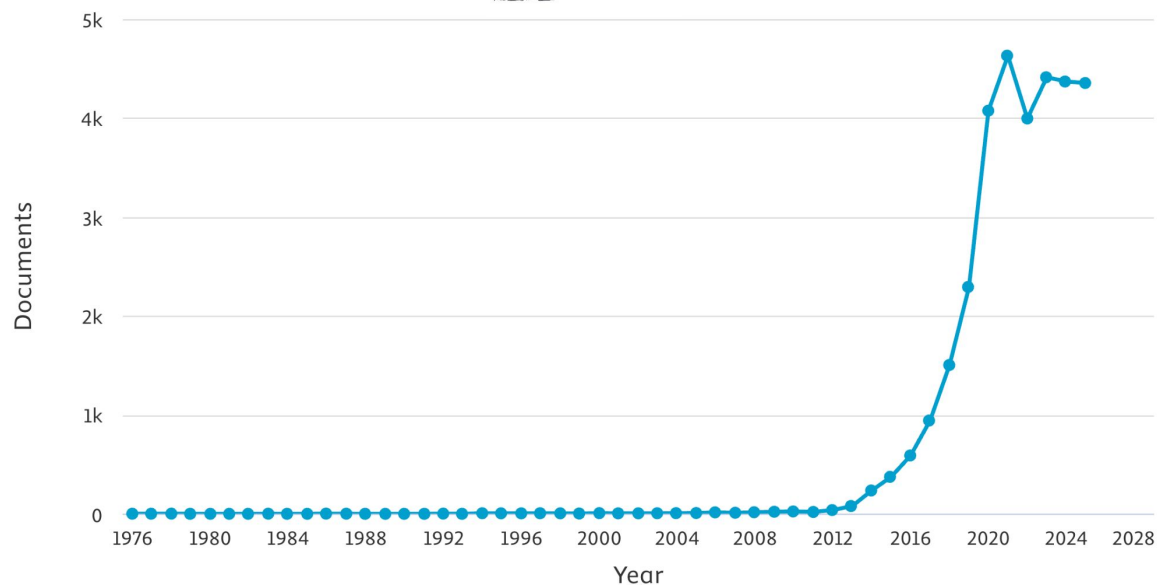


Publicaciones científicas

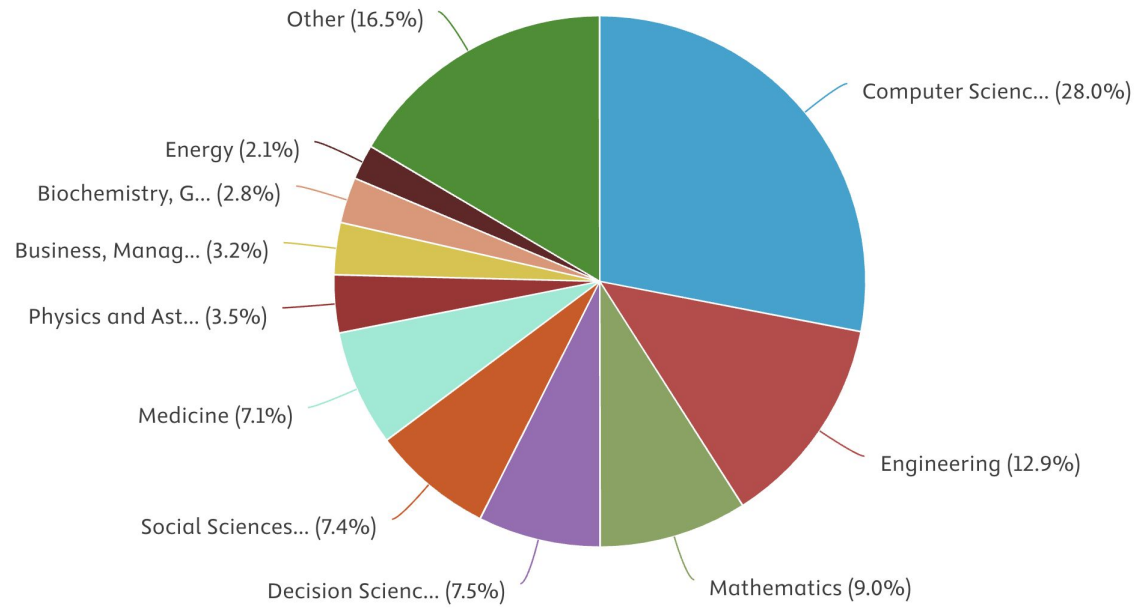
Documents by year



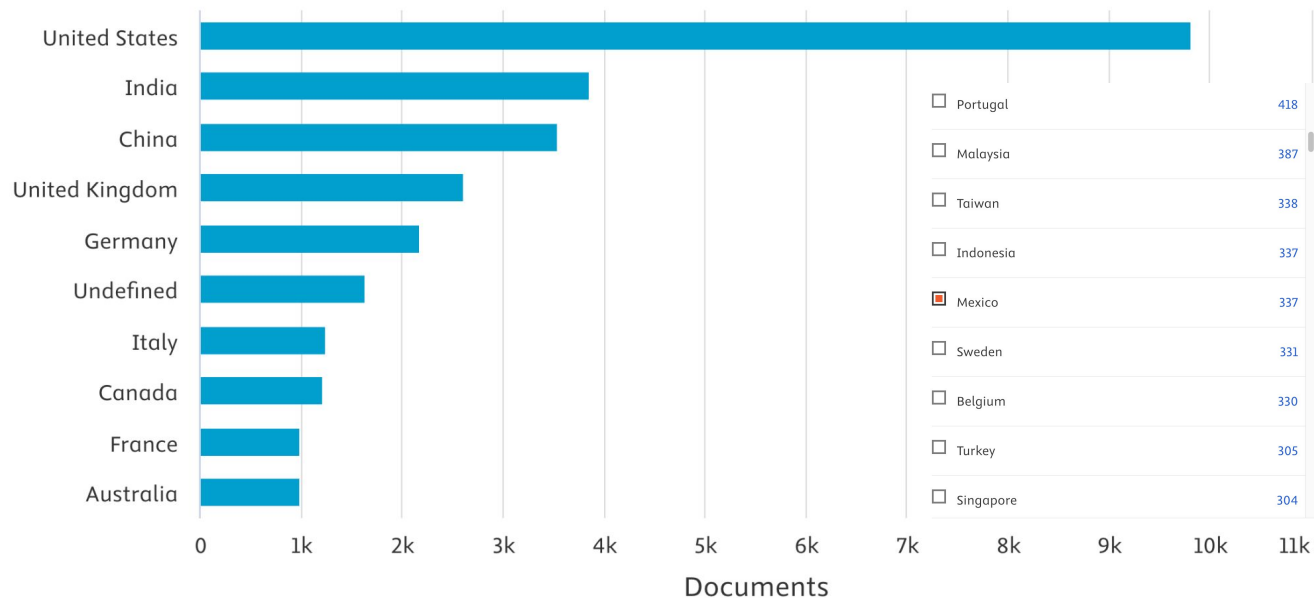
Scopus



Publicaciones por área



Publicaciones por país





What we are told...

DATA



SORTED



ARRANGED



PRESENTED
VISUALLY



EXPLAINED
WITH A STORY



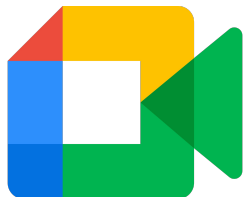
ACTIONABLE
(USEFUL)



What we actually have...



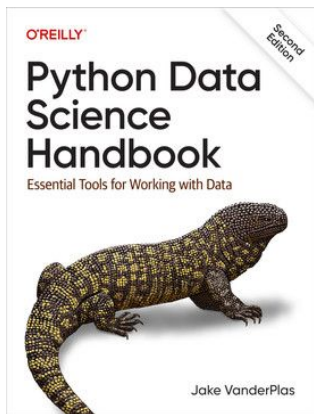
Recursos para el curso



Código de clase

lzl7cv22

Actividad asíncrona personal



- 00.00-Preface.ipynb
- 01.00-IPython-Beyond-Normal-Python.ipynb
- 01.01-Help-And-Documentation.ipynb
- 01.02-Shell-Keyboard-Shortcuts.ipynb
- 01.03-Magic-Commands.ipynb
- 01.04-Input-Output-History.ipynb
- 01.05-IPython-And-Shell-Commands.ipynb
- 01.06-Errors-and-Debugging.ipynb
- 01.07-Timing-and-Profiling.ipynb
- 01.08-More-IPython-Resources.ipynb

hussein@cicese.mx

